STPM 2023 PHYSICS 960/2

Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $m_p = 1.67 \times 10^{-27}$ kg, $k = \frac{1}{4\pi\varepsilon_0} = 9.0 \times 10^9$ N m^2 C^{-2}.

Section A [15 marks]

1. A cubic container contains three charges, $+Q$, $+4Q$ and $-3Q$. What is the total electric flux that passes through the wall of the container?

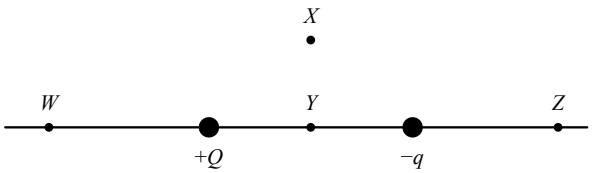
A 0 B $\frac{2Q}{\varepsilon_o}$ C $\frac{4Q}{\varepsilon_o}$ D $\frac{8Q}{\varepsilon_o}$

Answer: B

Gauss's law: $\Phi = \frac{Q_{\text{enclosed}}}{\varepsilon_o} = \frac{Q + 4Q - 3Q}{\varepsilon_o} = \frac{2Q}{\varepsilon_o}$

Explanation: The total flux depends only on the net charge enclosed, not on the shape of the container.

2. Two point charges, $+Q$ and $-q$, are separated by a certain distance as shown in the diagram below.



If the magnitude of Q is greater than that of q, at which position the electric field strength from these two charges may be zero?

A W B X C Y D Z

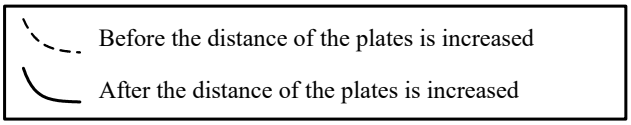
Answer: D

Between the charges the two fields point the same way, so they cannot cancel (not X or Y).

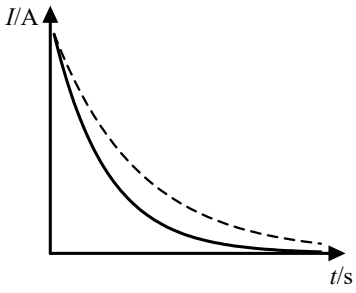
Outside, the fields are opposite; they can cancel only nearer the smaller charge $-q$, at Z.

Explanation: The larger charge must be further away for its field to be reduced to equal the smaller one's.

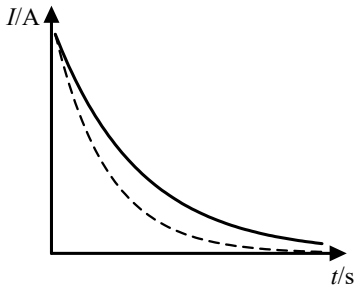
3. A parallel plate capacitor is connected to a battery and a resistor for charging process. If the distance between the plates is increased, which graph shows the variation of the current flow, I, with charging time, t, before and after the distance of the plates is increased for the capacitor?



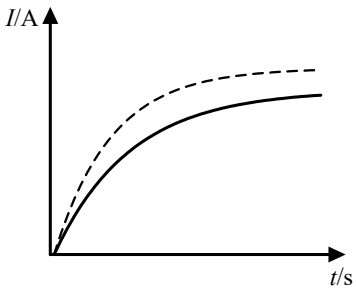
A



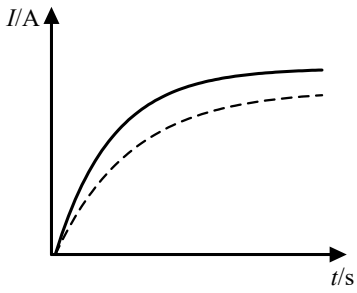
B



C



D



Answer: A

Increasing the plate separation reduces C, so the time constant RC decreases.

The initial current V/R is the same, but the current decays faster: the solid curve starts at the same value and falls more steeply (A).

Explanation: The charging current always decreases exponentially from V/R.

4. A parallel plate capacitor is initially charged by a battery and the battery is then disconnected. Which is true about the total charge and the energy stored in the capacitor if the distance between the plates is increased?

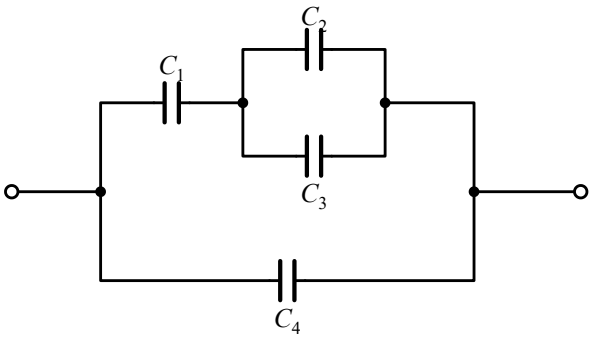
	Total charge	*Energy stored*
A	Remains constant	Decreases
B	Remains constant	Increases
C	Decreases	Decreases
D	Increases	Remains constant

Answer: B

With the battery disconnected the charge stays constant. Increasing d decreases C, so $U = \frac{Q^2}{2C}$ increases.

Explanation: The extra energy comes from the work done pulling the plates apart.

5. Four identical capacitors, C_1, C_2, C_3 and C_4, are arranged and connected to a power supply as shown in the circuit diagram below.



Which statement is true about the circuit?

A All capacitors have equal total charges.
B The total capacitance of C_2 and C_3 is equal to the capacitance of C_1.
C The total charge across C_1, C_2 and C_3 is equal to the total charge across C_4.
D The potential difference across C_1 is lower than the potential difference across C_4.

Answer: D

$C_2 \parallel C_3 = 2C$, in series with C_1: the p.d. divides so that $V_1 = \frac{2}{3}V$.

C_4 is directly across the supply: $V_4 = V$. So $V_1 < V_4$ (D).

Explanation: In series the smaller capacitance takes the larger share of the p.d.

6. A copper cable is used to supply current of 100 A. If the free electron density of copper is 8.00×10^{28} m^{-3} and the drift velocity of the free electron is 4.00×10^{-4} m s^{-1}, what is the radius of the cable?

A 1.25 mm B 2.50 mm C 4.42 mm D 5.00 mm

Answer: B

$$A = \frac{I}{nev_d} = \frac{100}{(8.00 \times 10^{28})(1.60 \times 10^{-19})(4.00 \times 10^{-4})} = 1.95 \times 10^{-5}\ \text{m}^2$$

$$r = \sqrt{(\frac{A}{\pi})} = 2.50 \times 10^{-3}\ \text{m} = 2.50\ \text{mm}$$

Explanation: $I = nAev_d$.

7. The resistivity of a semiconductor decreases when the temperature increases because

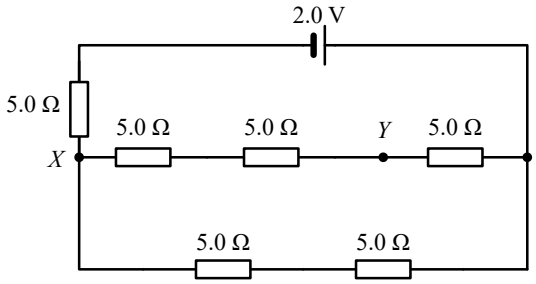
A the drift velocity of the charge carriers increases
B the kinetic energy of the charge carriers increases
C the number of charge carriers per unit volume increases
D the rate of collision between the charge carriers and atoms increases

Answer: C

In a semiconductor, heating frees more electrons (and holes): the number of charge carriers per unit volume increases greatly, so the resistivity falls.

Explanation: In a metal the carrier density is fixed, so increased collisions make the resistivity rise instead.

8. Six identical resistors of resistance 5.0 Ω are connected to a cell of e.m.f. 2.0 V of negligible internal resistance as shown in the circuit diagram below.



What is the potential difference between points *X* and *Y*?

A 0.73 V B 1.0 V C 1.1 V D 1.3 V

Answer: A

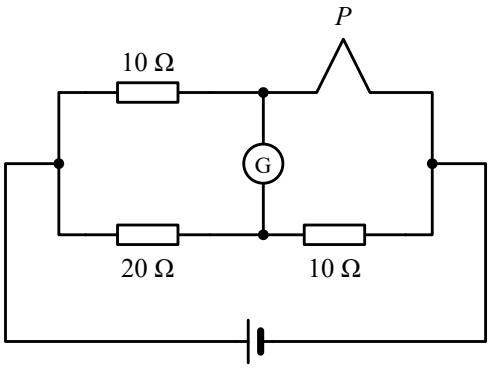
Middle branch (15 Ω) ‖ bottom branch (10 Ω) = 6.0 Ω; total = 6.0 + 5.0 = 11 Ω; $I = 2.0/11 = 0.182$ A.

p.d. across the parallel part = (0.182)(6.0) = 1.09 V; current in the middle branch = 1.09/15 = 0.0727 A

$$V_{XY} = (0.0727)(10) = 0.73 \text{ V}$$

Explanation: *X* and *Y* are separated by two 5.0 Ω resistors of the middle branch.

9. A resistive wire, *P*, of length 50 cm is connected in a Wheatstone bridge as shown in the circuit diagram below.



The cross-sectional area of *P* is 1.0×10^{-4} cm^2. If the galvanometer shows zero reading, what is the resistivity of *P*?

A 1.0×10^{-7} Ω m B 4.0×10^{-7} Ω m C 5.0×10^{8} Ω m D 1.0×10^{9} Ω m

Answer: A

$$\text{Balanced: } \frac{10}{20} = \frac{R_P}{10} \Longrightarrow R_P = 5.0\ \Omega$$

$$\rho = \frac{RA}{l} = \frac{(5.0)(1.0 \times 10^{-8})}{0.50} = 1.0 \times 10^{-7}\ \Omega\ \text{m}$$

Explanation: $1.0 \times 10^{-4}\ \text{cm}^2 = 1.0 \times 10^{-8}\ \text{m}^2$.

10. An ion of mass m and charge q moves perpendicularly into a uniform magnetic field at uniform velocity. Which statement describes the path of the ion in the magnetic field?

A A curve with the displacement from the initial linear path directly proportional to $\frac{q}{m}$

B A curve with the displacement from the initial linear path directly proportional to $\frac{m}{q}$

C A circle with its radius directly proportional to $\frac{q}{m}$

D A circle with its radius directly proportional to $\frac{m}{q}$

Answer: D

$$\text{The magnetic force gives circular motion with } r = \frac{mv}{qB} \propto \frac{m}{q}$$

Explanation: This is the basis of the mass spectrometer.

11. Which is **not** true about the application of Hall effect?

A To measure electric field

B To measure magnetic field

C To determine type of charge carriers

D To determine density of charge carriers

Answer: A

The Hall effect is used to measure magnetic fields and to find the type and density of charge carriers, not to measure an electric field. Statement A is not true.

Explanation: The Hall voltage is proportional to the magnetic flux density.

12. A coil of area 1.0 cm^2 with resistance 2.0 Ω and 100 turns is placed with the plane of the coil perpendicular to a magnetic field of 5.0 T. What is the average current that is induced in the coil when it is completely pulled out from the magnetic field in 0.50 s?

A 0.20 A B 0.10 A C 0.050 A D 0.0050 A

Answer: C

$$\varepsilon = \frac{NBA}{t} = \frac{(100)(5.0)(1.0 \times 10^{-4})}{0.50} = 0.10\ \text{V};\ I = \frac{0.10}{2.0} = 0.050\ \text{A}$$

Explanation: The flux linkage falls from NBA to zero.

13. Self-inductance of a solenoid can be increased by inserting a pure iron core because the iron core increases

A the current flow
B the magnetic flux density
C the resistance of the solenoid
D the mutual inductance between the iron core and the solenoid

Answer: B

Iron is ferromagnetic: it greatly increases the magnetic flux density (and so the flux linkage) for the same current, increasing L.

Explanation: $L = \mu N^2 A/l$, with $\mu \gg \mu_0$ for iron.

14. A resistor of resistance 100 Ω is connected in series to a sinusoidal voltage supply, $V = 220 \sin(100\pi t)$ where t is time in seconds. What is the average power dissipated in the resistor?

A 484 W B 342 W C 242 W D 171 W

Answer: C

$$\langle P \rangle = \frac{V_0^2}{2R} = \frac{220^2}{2(100)} = 242 \text{ W}$$

Explanation: The amplitude in the equation is the peak voltage.

15. A bulb which is labelled as 110 V, 55 W is connected in series with an inductor and a sinusoidal voltage supply of 220 V (r.m.s.) with frequency 50 Hz. What is the self-inductance of the inductor so that the bulb glows with normal brightness?

A 7.6 H B 1.4 H C 1.2 H D 0.70 H

Answer: C

$$I = \frac{55}{110} = 0.50 \text{ A}; R = \frac{110}{0.50} = 220 \text{ Ω}; Z = \frac{220}{0.50} = 440 \text{ Ω}$$

$$X_L = \sqrt{(440^2 - 220^2)} = 381 \text{ Ω} \Longrightarrow L = \frac{381}{2\pi(50)} = 1.2 \text{ H}$$

Explanation: The voltages across the bulb and the inductor are 90° out of phase, so they do not simply add to 220 V.

Section B [15 marks]

16. The electric field at the surface of a solid metal sphere is 3.0×10^6 N C^{-1}. The radius of the sphere is 20.0 cm.

(a) Calculate the charge on the sphere. [2]

Answer

$E = \dfrac{Q}{4\pi\varepsilon_0 r^2} \Rightarrow Q = \dfrac{(3.0 \times 10^6)(0.200)^2}{9.0 \times 10^9}$ 1

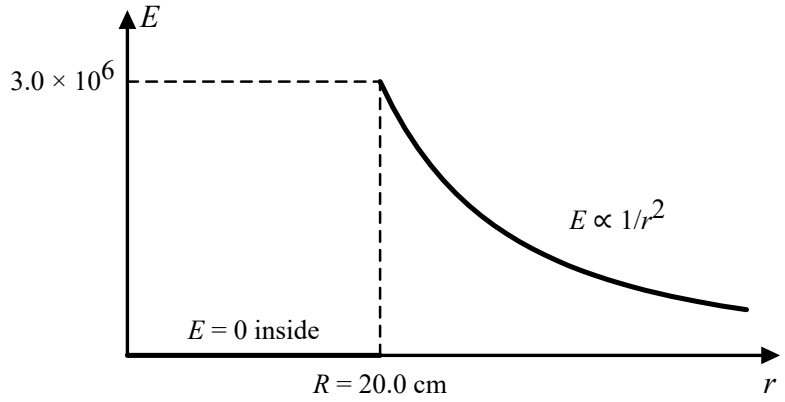
$Q = 1.33 \times 10^{-5}$ C 1

2

Explanation: Outside a charged sphere the field is as if all the charge were at the centre.

(b) Sketch the graph of electric field strength, E, against the distance, r, from the centre of the sphere. [2]

Answer



$E = 0$ inside the sphere ($r < R$) 1

Maximum at $r = R$, then decreasing as $1/r^2$ 1

2

Explanation: The charge on a conductor resides on its surface, so there is no field inside.

(c) If a stationary proton at the surface of the sphere is repelled from the sphere, calculate the speed of the proton at 40.0 cm from the centre of the sphere. [4]

Answer

$V = \dfrac{Q}{4\pi\varepsilon_0 r}$: $V(0.200) = 6.0 \times 10^5$ V, $V(0.400) = 3.0 \times 10^5$ V 1

Loss of potential energy = $e\Delta V = (1.60 \times 10^{-19})(3.0 \times 10^5)$ 1

$\dfrac{1}{2}mv^2 = e\Delta V \Rightarrow v = \sqrt{\left(\dfrac{2(1.60 \times 10^{-19})(3.0 \times 10^5)}{1.67 \times 10^{-27}}\right)}$ 1

$v = 7.58 \times 10^6$ m s^{-1} 1

4

Total for question **8**

Explanation: Energy conservation: electric potential energy lost = kinetic energy gained.

17. (a)

(i) State Kirchhoff's first law and Kirchhoff's second law. [2]

Answer

First law: the sum of currents entering a junction equals the sum of currents leaving it — 1

Second law: in any closed loop, the sum of e.m.f.s equals the sum of the products IR — 1

2

Explanation: They apply to any circuit.

(ii) State **one** fundamental principle that are associated with the Kirchhoff's laws. [1]

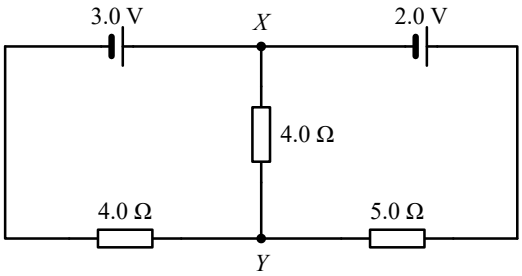
Answer

Conservation of charge (first law) / conservation of energy (second law) — 1

1

Explanation: Either principle.

(b) Two batteries and three resistors are connected as shown in the circuit diagram below.



Determine the current that flows through each battery. [4]

Answer

Let I_1 flow through the 3.0 V battery and I_2 through the 2.0 V battery; the 4.0 Ω resistor XY carries $I_1 + I_2$ — 1

Left loop: $3.0 = 4.0I_1 + 4.0(I_1 + I_2)$; right loop: $2.0 = 5.0I_2 + 4.0(I_1 + I_2)$ — 1

$8I_1 + 4I_2 = 3.0$ and $4I_1 + 9I_2 = 2.0$ — 1

$I_1 = 0.339$ A (3.0 V battery); $I_2 = 0.0714$ A (2.0 V battery) — 1

4

Total for question **7**

Explanation: Both batteries drive current through the shared 4.0 Ω resistor, which carries 0.411 A.

Section C [30 marks]

18. (a) A proton enters perpendicularly into a uniform magnetic field of 0.80 T with a velocity of 3.0×10^5 m s^{-1}.

(i) Determine the magnitude of acceleration of the proton. [4]

Answer

Magnetic force $F = evB$	1
$a = \frac{evB}{m}$	1
$a = \frac{(1.60 \times 10^{-19})(3.0 \times 10^5)(0.80)}{1.67 \times 10^{-27}}$	1
$a = 2.30 \times 10^{13}$ m s^{-2}	1
	4

Explanation: This is a centripetal acceleration; the speed does not change.

(ii) Determine the radius of the proton. [3]

Answer

$a = \frac{v^2}{r}$ (or $r = \frac{mv}{eB}$)	1
$r = \frac{(3.0 \times 10^5)^2}{2.30 \times 10^{13}}$	1
$r = 3.91 \times 10^{-3}$ m	1
	3

Explanation: Both methods give the same radius.

(iii) Describe the path of the proton if the magnetic field is directed into the page. [5]

Answer

The proton moves in a circle (arc) in the plane of the page	1
The magnetic force is always perpendicular to its velocity, so the speed is constant	1
With **B** into the page and the proton moving to the right, the force is initially upward (Fleming's left-hand rule)	1
So it curves anticlockwise	1
with constant radius 3.91 mm and period $2\pi m/eB$	1
	5

Explanation: $\mathbf{F} = e\mathbf{v} \times \mathbf{B}$.

(b) Describe the path of an electron that accelerates when the electron enters parallel to a uniform magnetic field. [3]

Answer

Moving parallel to **B**, the electron experiences no magnetic force (sin 0° = 0)	1
So its path is not deflected: it moves in a straight line along the field	1
Its velocity changes only because of other forces (e.g. an electric field), not because of **B**	1
	3
Total for question	**15**

Explanation: A magnetic field never changes the speed of a charge.

19. (a) State Faraday's law. [1]

Answer

The induced e.m.f. is directly proportional to the rate of change of magnetic flux linkage: $\varepsilon = -N\,\mathrm{d}\Phi/\mathrm{d}t$	1
	1

Explanation: The minus sign represents Lenz's law.

(b) Explain the phenomenon of self-induction. [3]

Answer

A changing current in a coil produces a changing magnetic flux linked with the coil itself	1
This induces an e.m.f. in the same coil (back e.m.f.)	1
which opposes the change in current (Lenz's law): $\varepsilon = -L\,\mathrm{d}I/\mathrm{d}t$	1
	3

Explanation: Self-induction delays the rise and fall of current in a coil.

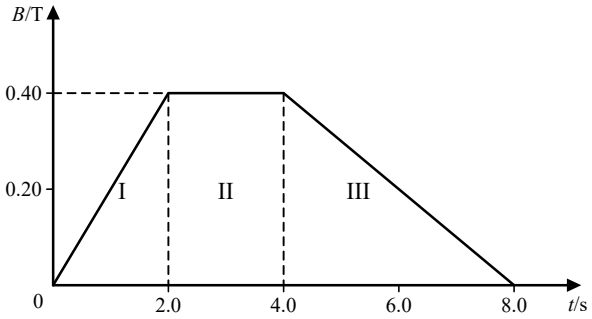
(c) State **three** ways to change the magnetic flux of a coil in a uniform magnetic field. [3]

Answer

• Change the magnetic flux density B	1
• Change the area of the coil in the field	1
• Change the angle between the coil and the field (rotate the coil)	1
	3

Explanation: $\Phi = BA\cos\theta$.

(d) A magnet bar is moved perpendicularly toward a single conducting loop in three stages, I, II and III, as shown in the graph of uniform magnetic field, B, against time, t, below.



(i) Calculate the induced e.m.f.s in stages I, II and III. [6]

Answer

$\varepsilon = A\frac{\Delta B}{\Delta t}$ (single loop of area A)	1
Stage I: $\frac{\Delta B}{\Delta t} = \frac{0.40}{2.0} = 0.20\ \text{T s}^{-1}$	1
$\varepsilon_{\text{I}} = 0.20A$ V	1
Stage II: B is constant	1
$\varepsilon_{\text{II}} = 0$	1
Stage III: $\frac{0.40}{4.0} = 0.10\ \text{T s}^{-1} \Longrightarrow \varepsilon_{\text{III}} = 0.10A$ V (opposite direction)	1
	6

Explanation: The e.m.f. is proportional to the gradient of the B–t graph. The area of the loop is not given, so the answers are in terms of A.

(ii) Determine the magnitude of an induced current in stage I if the resistance of the conducting loop is 5.0 Ω. [2]

Answer

$I = \frac{\varepsilon}{R} = \frac{0.20A}{5.0}$	1
$I = 0.040A$ ampere	1
	2
Total for question	**15**

Explanation: Ohm's law applied to the loop.

20. (a)

(i) What is meant by *alternating current*? [1]

Answer

A current that reverses its direction periodically (varies sinusoidally with time) — 1

1

Explanation: Its average value over a cycle is zero.

(ii) Explain **two** ways to increase the current in the *RC* circuit that has constant values of resistance, *R*, and capacitance, *C*. [4]

Answer

Increase the supply voltage — 1

$I = V/Z$, so a larger voltage gives a larger current — 1

Increase the frequency of the supply — 1

$X_C = 1/(2\pi fC)$ decreases, so the impedance falls and the current rises — 1

4

Explanation: *R* and *C* are fixed, so only *V* and *f* can change.

(b) A resistor of resistance 25 Ω and a capacitor of capacitance 100 μF are connected in series with a sinusoidal voltage supply which oscillates at a frequency of 60 Hz. The r.m.s. current of 1.25 A flows in the circuit.

(i) Calculate the peak voltage across the resistor and the capacitor. [6]

Answer

$I_0 = \sqrt{2}(1.25) = 1.77$ A — 1

$V_{R0} = I_0 R$ — 1

$= (1.77)(25) = 44.2$ V — 1

$X_C = \dfrac{1}{2\pi(60)(100 \times 10^{-6})} = 26.5\ \Omega$ — 1

$V_{C0} = I_0 X_C$ — 1

$= (1.77)(26.5) = 46.9$ V — 1

6

Explanation: These peaks do not occur at the same time; the supply peak is $\sqrt{(44.2^2 + 46.9^2)} = 64.4$ V.

(ii) Calculate the phase angle between the current and the voltage. [2]

Answer

$\tan \varphi = \frac{X_C}{R} = \frac{26.5}{25}$ 1

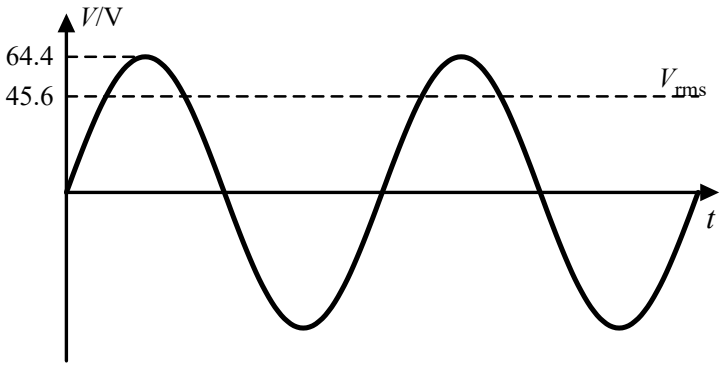
$\varphi = 46.7°$ (the current leads the voltage) 1

2

Explanation: In an *RC* circuit the current leads the supply voltage.

(iii) Sketch, on the same axis of time, the variation of r.m.s. voltage and the variation of voltage supply. [2]

Answer



Sinusoidal supply voltage (peak 64.4 V) and a constant r.m.s. line at 45.6 V 2

2

Total for question **15**

Explanation: The r.m.s. value is constant: it is peak ÷ √2.